

Presentation 1

1. Link to presentation and code on GitHub:

- Presentation Link: [Click Here](#) (Note: static visuals used in presentation; to see live dashboard use link below)
- Dashboard Link: [Click Here](#)
- Data cleaning code: [Click Here](#)

2. What did you do?

This week, my goal was to build off my previous visualization—which tracks call volumes by month—by adding hierarchical filtering by year and month. This way, Cynthia can easily compare this trend across the relevant time frames that she needs. To do so, I had to first download the new CAR data for the month of September. Then, building off of Krish's input code, I added a couple lines that ensure that we are only counting unique session IDs (similar to Julia's code). After this, I created a column that tracks the month of the call and a column that tracks the year of the call; these will be helpful when creating the hierarchical filter.

Next, I downloaded this cleaned monthly as a single .CSV and uploaded it to Power BI. Here, I created a line plot with the months in a calendar year as the x-axis and unique call session ID as the y-axis. I then added a hierarchical filter that easily allows the viewer to choose a specific year (either 2024 or 2025) and hone on a specific month(s).

While there did seem to be a slight increase in calls during summer 2024 and during the near year 2025, overall there did not appear to be any substantial seasonal trend that occurred in both years. However, I did notice that there have been less calls per month for most of 2025 compared to 2024. On average, it looks like about one to two thousand less calls per month, which would add up to twelve to twenty-four thousand calls a year. Though I have not been able to make any concrete recommendations based on this, I have a hypothesis moving forward that I want to test: if this drop off can be accounted for by certain issues being either resolved or more likely people feel comfortable reaching out for help due to today's political and social climate.

Moving forward, I hope to understand why there was this call drop off and if certain EP names (i.e. immigration, HIV, etc.) are responsible for this.

3. How does it help the project?

Understanding when call volumes peak is crucial for improving the efficiency of the call system, especially given the constraints legal aid organizations face. With limited budgets and unpredictable funding due to shifting policies and economic conditions, Legal Aid is already struggling to meet demand. Calls often exceed capacity, leaving people unable to get through or facing long wait times. Since demand likely fluctuates rather than remaining constant—spiking during particular days, weeks, or policy changes—identifying these seasonal or situational trends allows the organization to anticipate and prepare for them. Though more analysis is needed, it currently looks like call volumes on track to be less for 2025 compared to 2024. We need to understand why this is occurring before making any concrete recommendations. However, I wonder if certain issues might be best served by other resources, which would free up agents so that the organization can serve more clients effectively without significantly increasing costs.

4. Issues faced, attempt to resolve issues, and resolved issues:

One issue I faced is that when I was originally trying to add a filter for EP name, there ended up only being three potential EP name options post-data cleaning. This does not seem right so it must be an issue related to how I cleaned my data. While I tried to resolve it and build out a visualization for this week, I was not able to get there. This upcoming week I will try and do so.

5. Next Steps

- Create new visualization to understand why the drop off in calls occurred in 2025;
this would plot call volume total for each EP name against months of the year
with the goal of understanding which reason for calling has declined the most
- Work closely with teammates analyzing other metrics to link time-trend findings
with their insights for a more complete understanding

6. References

- Input code from Krish
- Julia's data transformations in python and PowerBI
- Information from Cynthia during class with her